

Sri Krishna

Data Engineer

Email : krishnass1121@gmail.com

Ph: 704-258-3136

Professional Summary

- Expert **Data Engineer** with over a 9+ years of experience in developing and managing **large-scale** data infrastructures on **AWS** and **GCP**.
- Specialized in **architecting, implementing, and maintaining scalable, high-availability systems** for **telecom, finance, and e-commerce industries**.
- Proficient in leveraging **AWS EC2, S3, EMR, Lambda, Redshift, Glue, CloudWatch** and **GCP's Big Query, Dataflow, Pub/Sub, GKE** for enterprise-level cloud data engineering.
- Designed and optimized complex **ETL pipelines** using **AWS Glue, GCP Dataflow, Apache Airflow, and Terraform**, ensuring seamless **data ingestion, transformation, and reporting**.
- Implemented real-time data processing frameworks utilizing **Apache Spark Streaming, Kafka, Kinesis, Google Pub/Sub** and for high-throughput analytics.
- Proven expertise in developing predictive models and machine learning solutions using **AWS Sage Maker, GCP AI Platform** and **Spark MLlib**.
- Adept at data warehousing, SQL performance tuning and distributed data processing across **Redshift, BigQuery, Oracle, MySQL** and **NoSQL databases**.
- Developed robust data governance, access control, and security frameworks using **AWS IAM, GCP IAM, Kerberos authentication** and **encryption protocols**.
- Strong leadership skills, having led teams of data engineers in high-stakes project environments to deliver innovative, cloud-native data solutions.
- Agile and Scrum methodology adept, ensuring **fast-paced** and **efficient project delivery** in **cross-functional teams**.

Technical Skills:

- **Cloud Platforms:** AWS (EC2, S3, EMR, Lambda, Redshift, Glue, CloudWatch, IAM, SNS, SQS, Kinesis), Google Cloud Platform (BigQuery, Dataflow, Pub/Sub, GKE, GCS, Cloud Functions)
 - **Programming Languages:** Scala, Python(Pandas, NumPy, Matplotlib), SQL
 - **Big Data Technologies:** Apache Spark, Apache Kafka, PySpark, HDFS, Zookeeper, Hadoop, Hive, Pig, SQOOP, Oozie, ELK (Elasticsearch, Logstash, Kibana)
 - **ETL & Data Pipelines:** Apache Airflow, Informatica PowerCenter, SSIS, Dataflow, AWS Glue
 - **Infrastructure as Code:** Terraform, CloudFormation
 - **Monitoring & Security:** AWS CloudWatch, AWS CloudTrail, GCP Monitoring, IAM Roles & Policies
 - **CI/CD & DevOps Tools:** GitHub, Jenkins, Docker, Kubernetes, Ansible
 - **Reporting & BI Tools:** Tableau, PowerBI, Apache Superset
-

Professional Experience:

Data Engineer

Furniture.com, Remote

June 2023 - Present

- Architected and deployed a real-time analytics platform across **AWS** and **GCP** to facilitate inventory management and dynamic pricing, enabling instant price adjustments in response to fluctuations in customer demand and competitive market trends.
- Created and refined predictive models utilizing **Spark MLlib**, which were then implemented through the **GCP AI Platform**, enhancing the accuracy of customer behavior forecasts to drive personalized recommendations and more effective marketing campaigns.
- Built and managed scalable data pipelines using **Kafka**, **GCP Pub/Sub** and **AWS Kinesis**, ensuring high throughput and zero data loss in handling over 100 million events per day.

- Optimized **AWS Redshift** and **BigQuery** queries, reducing data processing costs by 40% while enhancing query performance by 60% for business intelligence teams.
 - Integrated **Apache Airflow** with **AWS Glue** for **automated ETL workflows**, enabling seamless data transformation for real-time e-commerce analytics.
 - Led a team of data engineers to migrate on-prem batch processing to serverless **cloud-based architectures**, reducing operational overhead and improving scalability.
 - Developed recommendation engines using Python and Spark MLlib(PySpark), improving customer engagement and sales.
 - Created data pipelines using **Apache Airflow** and **PySpark** to handle large e-commerce datasets for personalized marketing.
 - Developed a recommendation engine using **TensorFlow on GCP AI Platform**, increasing conversion rates by 15%.
 - Implemented **AWS Lambda** functions for event-driven ETL processing, reducing data processing time by 50%.
 - Designed data visualization dashboards in **Tableau** and **PowerBI**, integrating with **AWS Athena** and **BigQuery** for real-time business intelligence reporting.
 - Automated cloud infrastructure deployment using **Terraform** and **CloudFormation**, ensuring 100% infrastructure-as-code consistency.
 - Enhanced data security by implementing **AWS IAM policies** and **GCP Identity-Aware Proxy (IAP)** for fine-grained access control.
-

Data Engineer

T-Mobile, Atlanta, GA

Oct 2019 - MAY 2023

- Developed a fraud detection system on AWS using **Spark Streaming** and **Kafka**, analyzing millions of transactions daily for real-time fraud detection and prevention.
- Implemented a centralized data lake on **AWS S3** and **GCP BigQuery**, enabling faster data access for customer analytics and network performance monitoring.

- Automated ETL processes using **AWS Glue** and **GCP Dataflow**, improving data ingestion efficiency and reducing operational costs.
 - Designed event-driven architectures using **AWS Lambda** and **Kinesis Firehose**, ensuring **real-time alerting** and **anomaly detection** for network failures.
 - Built scalable ML pipelines on **AWS SageMaker** and **GCP Vertex AI**, enabling predictive maintenance and network performance optimization.
 - Enhanced customer segmentation using **AI models**, resulting in a 20% increase in personalized marketing engagement.
 - Developed data processing scripts using Python and Pandas to handle IoT data streams, enabling real-time network health analysis.
 - Used PySpark for real-time data streaming and analysis from Kafka, supporting network performance monitoring.
 - Integrated **IoT data** from connected devices into AWS and GCP platforms, supporting real-time network health analysis.
 - **Deployed Elasticsearch** and **Kibana dashboards** for real-time log analysis, reducing incident response times by 30%.
-

Data Engineer

Athene, Des Moines, Iowa

Nov 2016 – Aug 2019

- Developed Spark-based real-time insurance risk assessment models, improving fraud detection and underwriting efficiency.
- Built data ingestion pipelines using **Apache Airflow** and **AWS Glue**, enhancing **data integrity** and accuracy for actuarial models.
- Implemented **ETL automation for policy data migration to AWS Redshift**, reducing manual processing efforts by 50%.
- Implemented data warehouse optimization efforts using **BigQuery**, improving policy lookup speeds by 40%.

- Created scalable **API-driven data pipelines** using **AWS Lambda** and **GCP Cloud Functions**, improving policyholder data accessibility.
 - Worked on **Kerberos authentication** and **AWS IAM role-based** access control, securing PII and financial data.
 - Designed **Terraform** scripts for cloud infrastructure automation, ensuring high availability and security compliance.
-

Database Developer

Infosys, Hyderabad, India

Aug 2015 – July 2016

- Developed SQL-based **ETL** processes for large-scale enterprise database migrations, ensuring seamless **data integrity**.
 - Optimized database query performance by implementing **indexing** and **partitioning strategies**, improving response times by 35%.
 - Automated data extraction and transformation processes using **Pythons scripts** with **Pandas** and **Numpy**, reducing data processing time by 45%
 - Built automated reporting solutions in **SQL Server** and **Oracle**, enabling real-time business intelligence for decision-makers.
 - Created **AWS RDS-based** high-availability database solutions, ensuring disaster recovery compliance.
 - Led a team in designing **NoSQL data** models for **MongoDB**, supporting unstructured data processing.
 - Developed **Python-based data ingestion scripts**, improving operational efficiency in data processing.
-

Education

- **Bachelor of Science in Electrical and Electronics Engineering**, Jawaharlal Nehru Technological University, India